

Nandana Dileep

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Experience

AI Engineer, JLR Technology & Business Services India July 2024 - Present

- Building **SARVAM**, JLR's Git-native Everything-as-Code platform, consolidating requirements, architecture, design, and tests in GitLab in place of DOORS/TRM/JIRA; built CLI, VS Code, and web tools for authoring, diagram previews, and traceability views.
- Implemented ISO 26262/ASPICE-aligned CI/CD gates for traceability, MISRA, and safety approvals; built the knowledge graph for **100% bidirectional requirement-to-test traceability**, orphan detection, and cross-repo ID uniqueness via a Bloom-filter registry.
- Deployed NVIDIA RAG Blueprint v2.6.2 on air-gapped AWS EKS with seven offline NIM models, including Llama 3.1 8B and Nemotron document extraction, for requirements Q&A, impact analysis, and traceability insights.
- Built and maintaining **Octopus**, which prioritizes tests using past results and code changes; adopted across JLR.
- Built and maintaining an ensemble classifier for 500K+ daily telemetry events, reducing test-rig costs by **18%**.
- Built a C++ GNSS processing pipeline with GTSAM for vehicle positioning; presented at the JLR Tech Symposium.

Software Development Engineer Intern, Amazon May 2023 - July 2023

- Designed and deployed a latency analysis service for Alexa using AWS Lambda, Coral, and OpenSearch.
- Built an end-to-end testing framework using JUnit 5 and Mockito.

Selected Projects

Mycelium - Reliable Tool Execution for AI Agents [GitHub](#) | [PyPI](#)

- Maintain an open-source Python runtime for agent tool execution, with guards against duplicate actions, unsafe retries, stale context, and invalid tool calls.
- Supports LangGraph, CrewAI, and custom Python agents; **30k downloads on PyPI**.

Meridian - AI Hiring Platform JLR Global Techathon Winner, 2026

- Built a platform that compares job descriptions with resumes, maps skill gaps for recruiter review, and delivers adaptive technical assessments.

LLM Serving Engine on Apple Silicon [GitHub](#)

- Built a local LLM serving engine with MLX, continuous batching, and PagedAttention-inspired KV cache management.
- Implemented OpenAI-compatible chat completions and published benchmarks against vLLM and llama.cpp.

Technical Skills

Languages: Python, C++, TypeScript, SQL

AI / ML: PyTorch, Transformers, LangGraph, CrewAI, LlamaIndex, RAG, MCP, MLX, scikit-learn

Backend / Data: FastAPI, Flask, PostgreSQL, Redis, Neo4j, OpenSearch, FAISS

Engineering: AWS, Docker, Git, Linux, Pytest, React, Next.js, Vercel AI SDK

Education

Indian Institute of Technology Madras Aug 2020 - May 2024
Bachelor of Technology, Civil Engineering

Selected Writing

[Mycelium: Turning AI Agents into Reliable Systems](#) | [LLM Serving with MLX](#)